Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 1 – Matrices

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. Find $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ -1 & 2 \end{pmatrix}$. **[2]**

2. Find $3\begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$. **[2]**

3. Find $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$. **[3]**

4. Find $\det\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$. **[2]**

5. Find the inverse of $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$. **[4]**

6. Find $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. **[2]**

7. State when a 2×2 matrix is singular. **[2]**

8. Find $\det\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$. **[4]**

9. Solve $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$. **[4]**

10. Find A^2 if $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. **[3]**

11. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, find A^T. **[2]**

12. Show that $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ is singular. **[2]**

13. Find the inverse of $\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$. **[3]**

14. Calculate $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. **[3]**

15. State the identity matrix of order 2. **[1]**

Section B – Hard (Q16–25)

16. Find the inverse of $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. **[6]**

17. Solve the system $x + 2y = 3$, $3x + 4y = 5$ using matrices. **[4]**

18. If $A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, find A^n. **[5]**

19. Show that $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ satisfies $A^2 = -I$. **[3]**

20. Find all 2×2 matrices that commute with $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. **[5]**

21. Given $AB = I$, show that $BA = I$ for square matrices of the same order (or discuss). **[4]**

22. Find $\det(A)$ if $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$. **[2]**

23. Use matrices to solve $2x + y - z = 1$, $x - y + z = 2$, $x + y + z = 3$. **[6]**

24. If $A^{-1} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, find A. **[3]**

25. Prove that $\det(AB) = \det(A)\det(B)$ for 2×2 matrices. **[5]**

End of Worksheet • *Show all working clearly.*